



NORTH SYDNEY BOYS HIGH SCHOOL

2009 YEAR 12 HSC ASSESSMENT TASK 1

Mathematics

General Instructions

- Working time – 65 minutes
- Write on one side of the paper (with lines) in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

Total Marks (59)

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Barrett
- ☐ Mr Fletcher
- ☐ Mr Ireland
- ☐ Mr Lowe
- ☐ Mr Rezcallah
- ☐ Mr Trenwith
- ☐ Mr Weiss

Student Number:

Question	1	2	3	4	5	6	Total	Total
Mark	$\frac{8}{8}$	$\frac{9}{9}$	$\frac{10}{10}$	$\frac{10}{10}$	$\frac{12}{12}$	$\frac{10}{10}$	$\frac{59}{59}$	$\frac{100}{100}$

Start each question on a new page.

Question 1 (8 marks) Start a new page.

Marks

- (a) Differentiate with respect to x :
- (i) $(x^2 + 3x)^4$ 2
- (ii) $\frac{x^2+1}{x-2}$ 2
- (b) (i) Find the x coordinate of the point on the graph of $y = 5 - 14x - 2x^2$ where the tangent is parallel to the line $y = -2x + 7$ 2
- (ii) Hence find the equation of this tangent. 2

Question 2 (9 marks) Start a new page.

- (a) In a certain arithmetic series, the second term is 19 and the eighth term is 37.
- (i) Show that the common difference is 3 2
- (ii) Find the value of the 51st term. 2
- (b) Evaluate $\sum_{k=2}^5 (k-1)^2$ 2
- (c) By considering the recurring decimal $0.4\dot{5}$ as the **sum of an infinite geometric series**, express $0.4\dot{5}$ in the form $\frac{a}{b}$. 3

Question 3 (10 marks) Start a new page.

Consider the curve given by $y = 1 + 3x - x^3$, for $-2 \leq x \leq 3$.

- (a) Find the coordinates of the stationary points and determine their nature. 4
- (b) Find the coordinates of any points of inflexion. 2
- (c) Sketch the curve in the domain $-2 \leq x \leq 3$. 3
- (d) What is the minimum value of the function in the domain $-2 \leq x \leq 3$? 1

~~SECRET~~

Start each question on a new page.

Question 4 (10 marks) Start a new page.

- (a) The current i in a certain resistor as a function of the power P developed in the resistor is given by $i = 2.6\sqrt{P}$.
Find the rate of change of i with respect to P when $P = 4$. 2
- (b) A particle moves in a straight line and after t seconds its velocity v metres per second given by $v = 12t - 3t^2$.
(i) When is the particle at rest? 2
(ii) What is the acceleration at $t = 2$? 2
- (c) Find all values of x for which the function $f(x) = 8x^2 - 24x + 5$ is increasing. 2
- (d) A curve $y = f(x)$ has the following properties:
 $f(x) > 0$; $f'(x) > 0$; $f''(x) < 0$.
Sketch a curve satisfying these conditions. 2

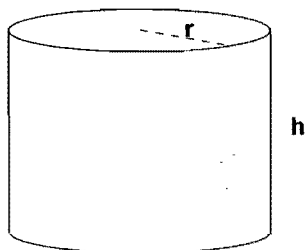
Question 5 (12 marks) Start a new page.

- (a) The first term of a geometric series is 7 and the 6th term is 1 701.
(i) Find the common ratio. 2
(ii) Calculate the sum of the first ten terms. 2
- (b) The sum S_n of the first n terms of a certain series is $2n + 3n^2$, for $n \geq 1$.
Find an expression for the n th term T_n of this series. 3
- (c) Mick received 30 tonnes of topsoil for his front yard. He uses a wheel barrow which can hold 150kg to spread the soil.
(i) How many loads in the wheel barrow will he use? 1
He begins at the pile of topsoil and deposits the first load 3 metres from the pile. Each successive load is dumped half a metre further from the pile, in a straight line.
(ii) How far from the pile will he leave the final barrow load? 2
(iii) What is the total distance that Mick will travel with the wheel barrow if he finishes back at his starting point? 2

Start each question on a new page.

Question 6 (10 marks) Start a new page.

- (a) The Green group at NSBHS wishes to build a new water tank. It is to be in the shape of a closed cylinder of radius r metres and height h metres, as shown in the diagram:



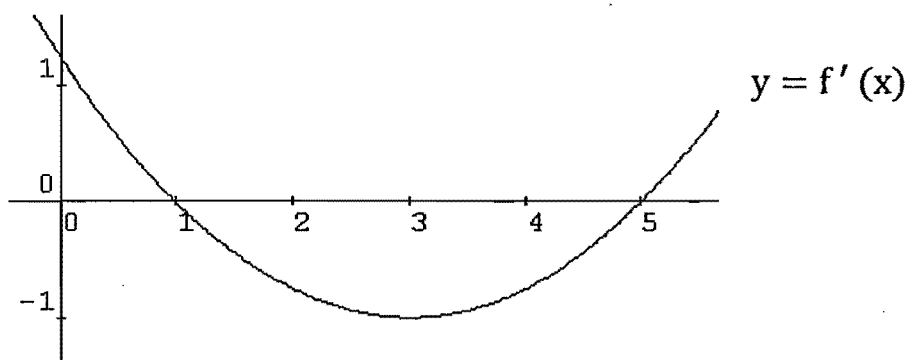
- (i) The surface area of metal to be used in the tank is 30m^2 .

Use that fact to show that $h = \frac{15}{\pi r} - r$ 2

- (ii) Show that the volume, $V\text{m}^3$ of the tank is given by $V = 15r - \pi r^3$ 1

- (iii) Find the radius if the volume of the tank is to be maximised. 3

- (b) The following diagram shows the graph of the gradient function of $f(x)$. For what value of x does $f(x)$ have a local minimum? Justify your answer. 2



- (c) By referring to the derivative of the function $f(x) = \sqrt{x-1}$, explain why the curve $y = \sqrt{x-1}$ has no turning points. 2

Q1. (a) (i) $\frac{d}{dx} (x^2+3x)^4 = (2x+3) \cdot 4 \cdot (x^2+3x)^3$ ✓
 $= (8x+12)(x^2+3x)^3$ ✓

(ii) $\frac{d}{dx} \left(\frac{x^2+1}{x-2} \right) = \frac{(x-2) \cdot (2x) - (x^2+1)(1)}{(x-2)^2}$ ✓

$= \frac{x^2 - 4x - 1}{(x-2)^2}$ ✓

(b) (i) $y = 5 - 14x - 2x^2$

$\therefore y' = -14x - 4x$

Gradient of $y = -2x + 7$ is -2

$\therefore -14 - 4x = -2$

$\therefore x = -3$

(ii) at $x = -3$, $y = 5 - 14(-3) - 2(-3)^2$ ✓
 $\therefore y = 29$

$\therefore y - 29 = -2(x + 3)$

$\therefore y = -2x + 23$

(i.e. $2x + y - 23 = 0$) ✓ for equation

✓ correctly equating

✓ correct answer

✓ for y value

✓ for equation

Q2.

(a) (i) $T_2 = a + d = 19$ ①

$T_8 = a + 7d = 37$ ----②

①-②→

$$6d = 18$$

$$d = 3$$

(ii) $T_{51} = a + 50d$

From (i), $a = 19 - d = 16$

$$\therefore T_{51} = 16 + 50(3)$$

$$\therefore T_{51} = 166$$

(b) $\sum_{k=2}^5 (k-1)^2 = 1^2 + 2^2 + 3^2 + 4^2$
 $= 1 + 4 + 9 + 16$
 $= 30$

(c) $0.4\dot{5} = 0.4 + 0.05 + 0.005 + 0.0005 + \dots$

G.P. with $a = 0.05$
 $r = 0.1$

$$\therefore S_{\infty} = \frac{0.05}{1-0.1}$$
$$= \frac{5}{90}$$

$$\therefore 0.4\dot{5} = \frac{4}{10} + \frac{5}{90}$$

$$= \frac{36+5}{90}$$

$$= \frac{41}{90}$$

} ✓ for working

✓ for $d=3$

✓ correct sub. in

✓ correct answer

✓ correct working

✓ correct answer

} ✓ correct G.P. parameters

✓ correct sub. into S_{∞}

✓ correct answer

Q3. $y = 1 + 3x - x^3$, $-2 \leq x \leq 3$

(a) For stationary points, $y' = 0$

$$\therefore y' = 3 - 3x^2 = 0$$

$$\therefore 3(1-x)(1+x) = 0$$

$$\therefore x = 1 \quad \text{or} \quad x = -1$$

$$y = 3 \quad \quad \quad y = -1$$

So stationary points are $(1, 3)$ and $(-1, -1)$

$$y'' = -6x$$

$\begin{cases} \text{At } (1, 3), y' = -6 < 0 \therefore (1, 3) \text{ is max. turning point} \\ \text{At } (-1, -1), y' = +6 > 0 \therefore (-1, -1) \text{ is min. turning point} \end{cases}$

[Alternatively, may use y' to test the points]

(b) For inflexion, $y'' = 0 \therefore -6x = 0$

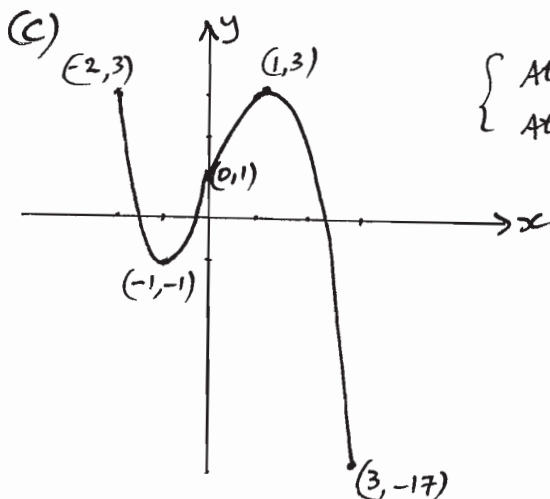
$$\therefore x = 0, y = 1$$

So inflexion could be $(0, 1)$

Test:

x	-0.1	0	$+0.1$
y''	$+$	0	$-$

Concavity changes, $\therefore (0, 1)$ is an inflexion pt.



$$\begin{cases} \text{At } x = -2, y = 1 - 6 + 8 = 3 \\ \text{At } x = 3, y = 1 + 9 - 27 = -17 \end{cases}$$

(d) Minimum value in domain $-2 \leq x \leq 3$ is -17

✓ Correct derivat equal to 0

✓ Correct x-coordinates

✓ for testing both points

✓ for correct coordinates: nature

✓ for $(0, 1)$

✓ for testing

✓ correct shape

✓ end points

✓ for the 3 points found earlier

✓ for the correct minimum

Q4.

$$(a) \quad i = 2.6\sqrt{p} \quad \therefore \frac{di}{dp} = \frac{1}{2}(2.6)p^{-\frac{1}{2}} \\ = \frac{1.3}{\sqrt{p}}$$

$$\therefore \text{at } p=4, \quad \frac{di}{dp} = \frac{1.3}{\sqrt{4}} = 0.65$$

✓ correct derivative

✓ correct answer

(b)

(i) 'at rest' $\Rightarrow v=0$

$$\therefore 12t - 3t^2 = 0$$

$$3t(4-t) = 0$$

$\therefore$ at rest at $t=0$ & $t=4$ seconds

✓ setting $v=0$

✓ for both times

$$(ii) \quad a = \frac{dv}{dt} = 12 - 6t$$

$$\therefore \text{at } t=2, \quad a = 12 - 6(2)$$

$$\therefore a = 0$$

So acceleration = 0 at $t=2$

✓ correct acceleration equation

✓ correct acceleration value

$$(c) \quad f(x) = 8x^2 - 24x + 5$$

$$f'(x) = 16x - 24$$

$$f(x) \text{ increasing} \Rightarrow f'(x) > 0$$

$$\therefore 16x - 24 > 0$$

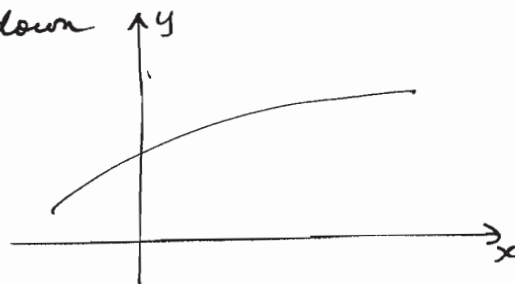
$$x > \frac{24}{16}$$

$$(i.e. \ x > \frac{3}{2})$$

✓ sets $f' > 0$

✓ correct answer

$$(d) \left\{ \begin{array}{l} f(x) > 0 \Rightarrow \text{above } x \text{ axis} \\ f'(x) > 0 \Rightarrow \text{increasing} \\ f''(x) < 0 \Rightarrow \text{concave down} \end{array} \right.$$



✓ correct possible curve

(deduct 1 error)

Q5.

(a) (i) $T_1 = a = 7$
 $T_6 = ar^5 = 1701$

$\therefore \frac{ar^5}{a} = \frac{1701}{7}$

$r^5 = 243$

$r = 3$

(ii) $S_{10} = \frac{7(3^{10} - 1)}{3 - 1}$ 9 means wrong formula - no marks
 $= 206\ 668$

✓ correct working.

✓ correct answer

✓ sub.in correctly

✓ correct answer

(b) $S_n = 2n + 3n^2$

$T_n = S_n - S_{n-1}$

$= (2n + 3n^2) - (2(n-1) + 3(n-1)^2)$

$= 2n + 3n^2 - (2n - 2 + 3n^2 - 6n + 3)$

$\therefore T_n = 6n - 1$

✓ sub.in n-1 correctly

✓ $S_n - S_{n-1}$ if they made an attempt to sub.in

✓ final answer

[alternatively, candidates may calculate successive sums $S_1, S_2, S_3 \dots$ and discern the pattern that way.]

- If they used this method, the needed to show it was an AP algebraically, i.e. not by showing first few terms have a common difference.

(2 marks for success)

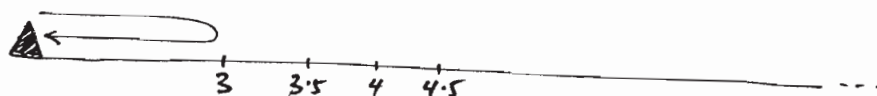
Q5 (c)

(i) number of loads = $\frac{30\,000}{150}$

$\therefore = 200 \text{ loads}$

✓ correct answer

(ii)



The distances from the pile form an A.P.

with $a=3$,
 $d=0.5$

$\therefore 200^{\text{th}} \text{ load} = T_{200} = a + (n-1)d$

$= 3 + 199(0.5)$

- wrong formula - no marks

$- S_n \quad - \text{no marks} = 102.5 \text{ m}$

Wrong

✓ correctly sub. into formula

✓ correct answer

(iii)



Total distance walked

$= 2(3 + 3.5 + 4 + \dots + 102.5)$

$= 2 \left[\frac{200}{2} (3 + 102.5) \right]$

$= 21.1 \text{ km}$

* wrong formula - no marks

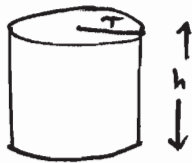
- T_n - no marks

✓ for calculate of S_n

✓ for include return trips

Q6.

(a)



(i)

$$\text{Surface area} = 2\pi r^2 + 2\pi r \cdot h$$

$$\therefore 2\pi r^2 + 2\pi r h = 30$$

$$2\pi r (r+h) = 30$$

$$r+h = \frac{30}{2\pi r} = \frac{15}{\pi r}$$

$$\therefore h = \frac{15}{\pi r} - r$$

✓ correct equation

✓ correctly simplified

(ii) $V = \pi r^2 \cdot h$

$$\therefore V = \pi r^2 \left(\frac{15}{\pi r} - r \right)$$

$$\therefore V = 15r - \pi r^3$$

✓ for this calculation

(iii) for max. volume, $\frac{dV}{dr} = 0$

$$\therefore 15 - 3\pi r^2 = 0$$

$$r^2 = \frac{5}{\pi}$$

$$\therefore r = \pm \sqrt{\frac{5}{\pi}}$$

but $r > 0 \therefore r = \sqrt{\frac{5}{\pi}}$

$$(\doteq 1.262 \text{ to 3 d.p.})$$

✓ correctly sets derivative = 0

✓ correct radius

Now $\frac{d^2V}{dr^2} = -6\pi r$

$$\therefore \text{at } r = \sqrt{\frac{5}{\pi}}, \frac{d^2V}{dr^2} < 0$$

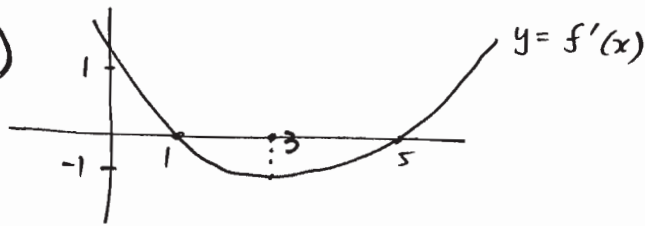
$$\therefore V \text{ is a maximum at } r = \sqrt{\frac{5}{\pi}}$$

✓ for testing correctly

[alternatively, test first derivative]

Q6.

(b)



Min. or max. occur when $f'(x) = 0$,
ie at $x = 1$ or $x = 5$.

For a minimum, $f'(x)$ must change
from $-$ to $+$.

$\therefore$ local minimum is at $x = 5$

✓ for $x = 5$
✓ for correct reasoning

(c) $f(x) = \sqrt{x-1}$

$$\therefore f'(x) = \frac{1}{2}(x-1)^{-\frac{1}{2}}$$

$$= \frac{1}{2\sqrt{x-1}}$$

But $\frac{1}{2\sqrt{x-1}}$ can never equal 0

$\therefore$ there are no turning points.

✓ for correct derivative
✓ for correct reasoning